FRAC-01 • SIDE A (PAGES 01–04) • BENCH FOLD: (1) FOLD BOTTOM HALF BEHIND TOP, (2) FOLD RIGHT HALF BEHIND LEFT SO PAGE 01 IS ON TOP

◀ FOLD 2: FOLD RIGHT BEHIND LEFT (01 ON COVER) | UNFOLD STEP 1: OPEN RIGHT ▶

01
PAGE / 08

FRACTION STUDIO PROPORTIONAL TOWERS

See & Feel Fractions: 4-Sided Blocks from 1 Whole to 1/12

01 02 03 04

SKU: FRAC-01 • KEEP IN LID

SIDE A • COVER (TOP-LEFT)

WHAT IS THIS BOX FOR? FRACTIONS WITHOUT THE TEARS (OR THE ROOT CANAL).

Every block in this box is scaled to a 1.0 Whole tower (84mm). Stack two towers side-by-side: if their tops are level, the fractions are equal!

4-SIDED BLOCKS: SPIN TO TRANSLATE

Why are 1/4, 0.25, and 25% taught in separate months of school?

- **Face 1: Fraction (e.g. 1/4)**
- **Face 2: Decimal (e.g. 0.25)**
- **Face 3: Percentage (e.g. 25%)**
- **Face 4: Calculator Reciprocal Key ([1/x])**

Just spin a block in your fingers to see all four languages at once!

THE 84MM "ONE WHOLE" RULE

Why 84 millimeters for 1 Whole? Because 84 divides cleanly by 2, 3, 4, 6, and 12! Every fraction column levels off at the exact same 84.00 mm ceiling.

WHAT'S INSIDE YOUR BOX (65 PIECES)

- 63 Fraction Blocks across 12 columns (10 denominators: 1/1 through 1/12)
- 1× 13U Socketed Storage Box + 1× Comparison Stage Lid
- Keep this folded guide inside your box lid for examples & part IDs!

★ YOUR FIRST 60 SECONDS WITH THIS BOX

START HERE!

1. SPIN THE PURPLE [1/4] BLOCK:
Rotate it 90° at a time in your fingers: read 1/4 → 0.25 → 25% → [4][1/x] on its four sides!

2. COMPARE TOWERS ON THE LID TRAY:
Stand [1/2], two [1/4]s, three [1/6]s, and six [1/12]s in the Lid Tray sockets—every tower is 42.0 mm!

3. PRESS THE BOX ICON KEY [FDISP]:
Add 1/3 + 1/4 on your calculator and press [FDISP] (your box lid icon!) to flip 0.5833 → 7/12.

FDISP

BOX ICON = CALCULATOR [FDISP] FRACTION↔DECIMAL TOGGLE KEY

The button icon embossed on your box lid (left) is the exact calculator key ([FDISP] & [g][SIMP])! Build every lesson with your blocks, then press the key.

4-SIDED BLOCKS: FRACTION ↔ DECIMAL ↔ [FDISP] BOX ICON

STEP A: HANDS-ON WITH BLOCKS ON LID TRAY

1:1 CAD SCENE

FRACTIONS
1/2
1/4
1/4
1/6
1/3

Face 1: Fraction	Face 2: Decimal	Face 4: Box Icon
1/2 + 1/4 + 1/4 = 1	0.50 + 0.25 + 0.25	[FDISP] & [f][1/x]

▲ **Spin any block 90° at a time to read Fraction, Decimal,** Percentage, and RPN Reciprocal keys!

FDISP

STEP B: TRY IT ON YOUR CALCULATOR

BOX ICON: [FDISP] & [g][SIMP

DECOMPOSITION TREE: 1 WHOLE = 1/2 + 1/4 + 1/4

1 WHOLE (84.0 mm)
1/2 (42 mm)
2/4 = 1/2 (42 mm)
0.50 (50%)
1/4 (0.25)
1/4 (0.25)

42 + 21 + 21 = 84.0 mm (100%)

PRESS BOX ICON & RPN KEYS (SC6):

1. ADD PARTS:
1 ENTER 2 ÷ + 1/4 + 1/4

2. BOX ICON KEY:
FDISP = 1 (Whole)

✓ DISPLAY (X) = 1

KEYSTROKES: 1 [ENTER] 2 [÷] 1 [ENTER] 4 [÷] [+] 1 [ENTER] 4 [÷] [+] [F

Build 1/2 + 1/4 + 1/4 on the Lid Tray (= 84mm Whole), then verify on the calculator with [FDISP]!

WELCOME TO HANDS-ON MATH • KEEP THIS GUIDE IN YOUR BOX LID

STEP 1: UNFOLD RIGHT → PAGE 02

02
PAGE / 08

HOW IT WORKS: THE CORE IDEA

True Proportional Heights & 4-Way Notation Faces

01 02 03 04

SKU: FRAC-01 • KEEP IN LID

SIDE A • 1ST UNFOLD (TOP-RIGHT)

INVERSE PROPORTIONALITY: BIGGER DENOMINATOR = SHORTER BLOCK

A fraction 1/n slices 1 whole unit into n equal pieces.
The visible height of every block is 84.00 mm divided by n:

$$h_{vis}(1/n) = H_{whole} / n = 84.00\text{ mm} / n$$

When you snap blocks into a tower, the bottom plug disappears inside the socket below it, so the heights add cleanly:

$$H_{tower} = 1.85\text{ mm} + \Sigma\,(84.00\text{ mm} / n_i)$$

HOW THE FRACTION SYSTEM FITS TOGETHER

- Whole Tower Height: H_whole = 84.00 mm (12U Unit Stacking Pitch)
- Base Unit Block: 1/12th block = 7.00 mm = 1U unit height
- Cross-Kit Harmony: 1/12 block matches Prime Block [2] (7.00mm)!
- Instant Check: Run your finger across two towers to feel if they are equal

FOUR LANGUAGES ON EVERY BLOCK

Switching between fractions, decimals, and percentages trips up almost everyone at first. Because all four faces are engraved around the same block, you see immediately that 1/4, 0.25, and 25% are the exact same physical amount wearing different outfits!

3 CLASSIC MATH TRAPS THIS KIT MAKES IMPOSSIBLE

VISUAL PROOF

✗ **TRAP 1 – Adding denominators (1/2 + 1/3 ≠ 2/5):**
✓ **FIX:** On the Lid Tray, [1/2] + [1/3] is 70mm (5/6)—over twice as tall as 2/5 (33.6mm)!

✗ **TRAP 2 – Thinking bigger denominator = bigger piece:**
✓ **FIX:** Seat [1/12] (7mm) next to Green [1/2] (42mm) on the Lid Tray: more slices = shorter blocks.

✗ **TRAP 3 – Treating decimals as separate math:**
✓ **FIX:** Spinning any block or pressing [FDISP] proves 0.25 and 1/4 are the exact same 21mm height.

CALCULATOR CHECK: 1 [ENTER] 2 [÷] 1 [ENTER] 4 [÷] [+] 1 [ENTER] 4 [÷] [+] → X = 1

10 CURRICULUM DENOMINATOR TIERS REFERENCE

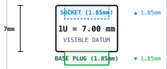
TIER	DECIMAL	PERCENT	HEIGHT	UNITS (U)
1 / 1	1.000	100.0%	84.00 mm	12.00 U
1 / 2	0.500	50.0%	42.00 mm	6.00 U
1 / 3	0.333	33.3%	28.00 mm	4.00 U
1 / 4	0.250	25.0%	21.00 mm	3.00 U
1 / 5	0.200	20.0%	16.80 mm	2.40 U
1 / 6	0.167	16.7%	14.00 mm	2.00 U
1 / 8	0.125	12.5%	10.50 mm	1.50 U
1 / 9	0.111	11.1%	9.33 mm	1.33 U
1 / 10	0.100	10.0%	8.40 mm	1.20 U
1 / 12	0.083	8.33%	7.00 mm	1.00 U
Box & Lid	13U Case	12-Lane	98.40 mm	13.00 U

WHY YOUR BRAIN LOVES THIS

When two fraction towers are level to your fingertips, you don't need to guess common denominators – you can physically see and feel why they match!

1:1 CAD REFERENCE VIEW

1:1 BLOCK CAD ANATOMY



✓ FLUSH PLUG-AND-SOCKET JOINT
Socket swallows plug completely:
Only 7.00mm true heights add!

BOX ICON & FORMULA QUICK-REFERENCE

Box Button Icon Key:
[FDISP] Fraction ↔ Decimal Toggle

Block Height Law:
H(1/n) = 84.00 mm ÷ n

Face 1 → Face 2 → Face 3:
1/n → Decimal → Percent (%)

Face 4 (Reciprocal):
Press [f] [1/x] (SC-6) or [1/x] (SC-32)

Simplify Fractions:
[g] [SIMP] reduces to lowest terms

PROPORTIONAL HEIGHT SYSTEM • H(1/N) = 84.0 MM ÷ N

STEP 2: UNFOLD DOWN ▼ PAGES 03–04

▲ FOLD 1: FOLD BOTTOM HALF BEHIND TOP ▲

▼ UNFOLD STEP 2: OPEN DOWN FOR PAGES 03 & 04 ▼

03
PAGE / 08

EXAMPLE 1: HANDS-ON WALKTHROUGH

Equal-Height Proof: Why 1/2 + 1/4 + 1/4 Equals 1 Whole

01 02 03 04

SKU: FRAC-01 • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT-LEFT)

① ESTABLISH 1.0 DATUM & EQUIVALENCE

1. SEAT THE 1.0 WHOLE REFERENCE TOWER:
Place the gold 84.00mm Whole block into the comparison socket.
Its summit establishes the benchmark ceiling for 1 whole unit.

2. BUILD EQUIVALENT TOWERS SIDE-BY-SIDE:

- Take two 1/2 blocks (42mm each): stack them → exactly 84.00 mm!
- Take one 1/2 block and two 1/4 blocks: 42 + 21 + 21 = 84.00 mm!
- Take four 1/4 blocks: 21 × 4 = 84.00 mm!

3. ROTATE MULTI-NOTATION FACES:
Turn blocks to verify decimal and percent sums match 1.0 and 100%:
0.50 + 0.25 + 0.25 = 1.00 | 50% + 25% + 25% = 100%

BENCHMARK LEVELING PROOF

Slide a straight-edge across the top of both towers. They meet seamlessly with zero step, zero gap, and zero wobble. The physical height proves mathematical equivalence directly to the student's fingertips.

FDISP

STEP B: TRY EXAMPLE #1 ON YOUR CALCULATOR

[FDISP] & [g][SIMP

STEP 1 (ENTER 4/6):
Press 4 [ENTER] 6 [÷] → Pushes 0.6667 (four Blue [1/6] blocks = 56.0 mm) onto the stack.

STEP 2 (ADD 1/12):
Press 1 [ENTER] 12 [÷] [+] → Adds 0.0833 (7.0 mm) to reach 0.7500 (63.0 mm total height).

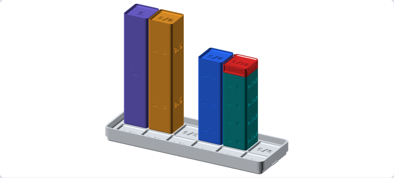
STEP 3 (PRESS BOX ICON [FDISP]):
Press [FDISP] → Converts 0.75 into exact reduced fraction 3/4 (matching three [1/4] blocks)!

LIVE 4-REGISTER RPN STACK READOUT (AFTER KEYSTROKES):
T: 0 | Z: 0 | Y: 0 | X: 3/4 → DISPLAY (X) = 3/4

LESSON 1: LID TRAY EQUIVALENCE ↔ [FDISP] KEYSTROKES

STEP A: HANDS-ON WITH BLOCKS ON LID TRAY

1:1 CAD SCENE



1.0 Whole Tower	3/4 Tower	4/6 + 1/12 Tower
84.00 mm (12U)	63.00 mm (9U)	63.00 mm (9U)

▲ **Seat fraction towers on the 12-socket Lid Tray: when two towers** stand level, the fractions are equal!

FDISP

STEP B: TRY IT ON YOUR CALCULATOR

BOX ICON: [FDISP] & [g][SIMP

EQUIVALENCE TREE: 4/6 + 1/12 = 3/4 (63.0 MM)

9/12 = 3/4 (63.0 mm)
4 × [1/6] (56mm)
+ 1×[1/12] (7mm)
4/6 = 8/12
+ 1/12
= 3/4 (63mm)

63.0 mm = Three [1/4] Blocks on Lid!

PRESS BOX ICON & RPN KEYS (SC6):

1. 4/6 + 1/12:
4 ENTER 6 ÷ 1 ENTER 12 ÷

2. BOX ICON KEY:
+ FDISP = 3/4

✓ DISPLAY (X) = 3/4

KEYSTROKES: 4 [ENTER] 6 [÷] 1 [ENTER] 12 [÷] [+] [FDISP]

Compare fraction towers on the Lid Tray, then press the Box Icon key [FDISP] for exact reduced fractions!

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 3/4 → X = 3/4

4 NOTATIONS ON EVERY BLOCK • FRACTION, DECIMAL, PERCENT & RPN

CONTINUE RIGHT → PAGE 04

04
PAGE / 08

EXAMPLE 2: CORE MATHEMATICAL PRINCIPLE

Adding Unlike Denominators: 1/3 + 1/4 = 7/12 by Leveling

01 02 03 04

SKU: FRAC-01 • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT-RIGHT)

② UNLIKE DENOMINATOR ADDITION: 1/3 + 1/4

1. THE CLASSIC FRACTION DILEMMA: ADD 1/3 + 1/4
Students frequently make the error of adding across: 1/3 + 1/4 ≠ 2/7.
Stack one 1/3 block (28.0mm) and one 1/4 block (21.0mm) together:
Combined Tower Height = 28.00 + 21.00 = 49.00 mm

2. FIND THE COMMON LEVELING MEASURING ROD:

- Can 1/6 blocks measure it? 14 × 3 = 42mm (gap), 14 × 4 = 56mm (overshoots!).
- Can 1/8 blocks measure it? 10.5 × 4 = 42mm, 10.5 × 5 = 52.5mm (overshoots!).
- **Test the 1/12 blocks (7.00 mm):**

Stack 1/12 blocks one by one alongside the tower...

✓ EXACTLY 7 TWELFTH BLOCKS LEVEL PERFECTLY!

7 × (1/12) = 7 × 7.00 mm = 49.00 mm

1/3 + 1/4 = 4/12 + 3/12 = 7/12

Four 1/12 blocks level with 1/3 (28mm). Three 1/12 blocks level with 1/4 (21mm). Together, 4 + 3 = 7 blocks level with the summit with zero remainder!

PHYSICAL COMMON DENOMINATOR PROOF

Students never have to memorize finding the LCD blindly again. The physical leveling proves why 12 is the common denominator and why the answer must be 7/12.

★ 2-PLAYER GAME: 'MYSTERY TOWER MATCH' (2 MINS)

Player A builds a 2-block tower on the Lid Tray using unlike denominators (e.g., [1/2] + [1/6] = 56 mm). Player B must find a single-color tower of matching blocks that stands at the exact same height **(two [1/3] blocks = 2/3!) and verify with [FDISP] on the calculator.**

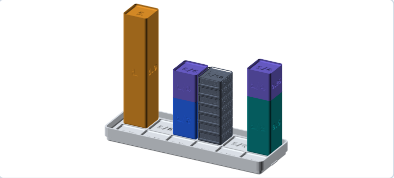
⚙ WHERE THIS SHOWS UP IN ENGINEERING, OPTICS & ELECTRONICS

Parallel resistors and camera lenses add by reciprocals: $1/R_{total} = 1/R_1 + 1/R_2$!
When you stack [1/3] + [1/4] = 7/12 on the Lid Tray and verify with [FDISP] on your calculator, **you are using the exact harmonic fraction math electrical engineers use every day.**

LESSON 2: 1/3 + 1/4 = 7/12 ON LID TRAY ↔ [FDISP] KEY

STEP A: HANDS-ON WITH BLOCKS ON LID TRAY

1:1 CAD SCENE



1/3 Block (Blue)	1/4 Block (Purple)	Seven 1/12 (Gray)
28.00 mm	21.00 mm	7 × 7mm = 49.00 mm

▲ **On the Lid Tray, [1/3] + [1/4] (28 + 21 = 49 mm) stands** dead-level with seven [1/12] blocks!

FDISP

STEP B: TRY IT ON YOUR CALCULATOR

BOX ICON: [FDISP] & [g][SIMP

COMMON DENOMINATOR TREE: 1/3 + 1/4 = 7/12

1/3 + 1/4 = 7/12 (49 mm)
1/3 = 4/12 (28mm)
1/4 = 3/12 (21mm)
4×[1/12]
+ 3×[1/12]
= 7/12

LCD(3,4) = 12 • 28 + 21 = 49.0 mm

PRESS BOX ICON & RPN KEYS (SC6):

1. 1/3 + 1/4:
1 ENTER 3 ÷ 1 ENTER 4 ÷

2. BOX ICON KEY:
+ FDISP = 7/12

✓ DISPLAY (X) = 7/12

KEYSTROKES: 1 [ENTER] 3 [÷] 1 [ENTER] 4 [÷] [+] [FDISP]

Pressing [FDISP] (the exact icon embossed on your Fraction Studio box!) toggles 0.5833 ↔ 7/12!

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 7/12 → X = 7/12

COMMON DENOMINATORS MADE VISIBLE • MATCH HEIGHTS TO ADD

STEP 3: FLIP SHEET OVER ↻ PAGE 05

◀ PAGE 03 (LEFT) → PAGE 04 (RIGHT) • THEN FLIP SHEET OVER FOR PAGES 05–08 ▶

◀ PAGE 05 (CHALLENGES) → PAGE 06 (HOW TO STACK BLOCKS BACK IN) ▶

EXAMPLE 3 & DESK CHALLENGES

Rotate Any Block to Translate Fractions, Decimals & Percents

3 INSTANT FRACTION ↔ DECIMAL ↔ PERCENT TRANSLATION

THE GOAL: Prove that 3/4, 0.75, and 75% are the exact same quantity.

1. STACK THREE 1/4 BLOCKS:
Snap three Matte Blue [1/4] blocks together on the Lid Tray (height = 63 mm).

2. SPIN THE TOWER 90° AT A TIME:
Face 1 reads 1/4 + 1/4 + 1/4 = 3/4. Spin 90°: Face 2 reads 0.25 × 3 = 0.75.

3. COMPARE AGAINST 3/4 OF A WHOLE:
Spin again: Face 3 reads 25% × 3 = 75%. Face 4 shows the RPN key [4][1/x]!

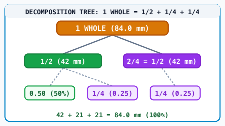
✓ FOUR LANGUAGES FOR ONE PHYSICAL REALITY

Fractions, decimals, and percents aren't three different kinds of math.
They are just four ways of naming the exact same 21 mm block height!
When students rotate a block in their hand, 1/5 = 0.20 = 20% becomes obvious.
Use Face 4 ([n][1/x]) with your SC-32 calculator to see live decimal reciprocals.

PRO TIP: CLOSE YOUR EYES & FEEL THE ANSWER

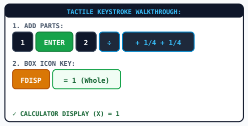
Once you build an example, run your fingers across the pieces. Your hands remember physical shapes and heights way longer than memorized rules!

EXAMPLE 3: VISUAL DECOMPOSITION TREE


DECOMPOSITION TREE: 1 WHOLE = 1/2 + 1/4 + 1/4

42 + 21 + 21 = 84.0 mm (100%)

STEP B: PRESS BOX ICON & RPN KEYS (SC6)


TACTILE KEYSTROKE WALKTHROUGH:

✓ CALCULATOR DISPLAY (X) = 1

FDISP
FULL KEYSTROKES: 1 [ENTER] 2 [÷] 1 [ENTER] 4 [÷] [+] 1 [ENTER] 4 [÷] [+] [FDISP]
✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 1 → X = 1

3 DESK CHALLENGES – TRY THESE NOW!

WARM-UP #1 **Build 4 different single-color towers that match the [1/2] block!**
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ **Build first, then flip booklet upside-down to check Answer #1 below!**

LEVEL-UP #2 **Stack [1/3] + [1/6]. Which single block is the exact same height?**
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ **Build first, then flip booklet upside-down to check Answer #2 below!**

BRAIN-MELTER #3 **Find the height gap between [3/4] and [1/3] using [1/12] blocks!**
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ **Build first, then flip booklet upside-down to check Answer #3 below!**

SELF-CHECK ANSWER KEY (ROTATE BOOKLET 180° TO READ)
#3: Five [1/12] blocks (5/12)
#1: 2/4, 3/6, 4/8, 6/12 | #2: [1/2] (28mm + 14mm = 42mm)

FOR PARENTS & TEACHERS (Grades 3–8 (Fractions, Decimals, Percents & Ratios))

- CCSS.MATH.CONTENT.3.NF.A.3: Explain equivalence of fractions and compare by size.
- CCSS.MATH.CONTENT.4.NF.C.6: Use decimal notation for fractions and compare quantities.
- CCSS.MATH.CONTENT.5.NF.A.1: Add and subtract fractions with unlike denominators.
- **CCSS.MATH.PRACTICE.MP2, MP4, MP5, MP7: Reason, Model, Use Tools & Structure.**

Next up on Page 06 → How to stack everything back in the box in 10 seconds!

BOSS CHALLENGE #4 **BLOCKS → BOX ICON KEY SPEED-RUN!**

1. Build Examples #1, #2, and #3 hands-on with your physical pieces.
2. Press the Box Button Icon ([FDISP] & [g][SIMP]) and keystrokes on your calculator:
- **1 [ENTER] 2 [÷] 1 [ENTER] 4 [÷] [+] 1 [ENTER] 4 [÷] [+] [FDISP] → X = 1**

CHECK OFF YOUR MASTERY MILESTONES:

☐ 1. Demonstrated Lessons #1, #2 & #3 hands-on with the physical blocks
☐ 2. Solved all 3 Desk Challenges above (check upside-down key if stuck!)
☐ 3. Demonstrated each lesson on the calculator using the Box Button Icon
☐ **4. Packed all pieces back into the box in under 10 seconds (see Page 06 →)**

TEST YOURSELF • ANSWERS INCLUDED SO YOU CAN CHECK RIGHT AWAY

NEXT: HOW TO PACK UP → PAGE 06

HOW TO STACK IT ALL BACK IN

The 10-Second Self-Checking Pack-Up Puzzle

TOP-DOWN BOX MAP: WHERE EVERY PIECE LIVES

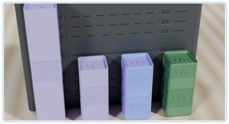
Putting the blocks away is a 10-second puzzle. Match each piece to its labeled home bay below so everything nests flush and the lid snaps shut:

1x Whole "1"	1x Whole "1"	2x Half "1/2"	2x Half "1/2"	3x Third "1/3"	4x Fourth "1/4"
5x Fifth "1/5"	6x Sixth "1/6"	8x Eighth "1/8"	9x Ninth "1/9"	10x Tenth "1/10"	12x Twelfth "1/12"

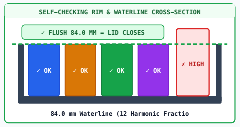
✓ THE 84.0 MM STORAGE BOX WATERLINE CHECK (▼)

The 13U Storage Box has 12 columns across 10 denominators (2×[1], 2×[1/2], plus [1/3] to [1/12]). Because N blocks of size 1/N always add up to 1 Whole (84.0 mm), every single column levels off at the exact same 84.0 mm line when all 63 blocks are home. If any column looks short, you can tell in 2 seconds which fraction is still on the table!

1. STORAGE CASE & COMPARTMENTS



2. WHY THE LID SELF-CHECKS EVERY PIECE


SELF-CHECKING RIM & WATERLINE CROSS-SECTION

84.0 mm Waterline (12 Harmonic Fractio

▲ **SELF-INDEXING BAYS & EMBOSSED BOX BUTTON ICON:**
Calibrated bays guide every piece home, and the lid icon links directly to your calculator.

THE 10-SECOND PACK-UP ROUTINE

HOW TO STACK EVERYTHING BACK IN:

1. SORT BY COLOR & NUMBER:
Group matching blocks or tiles together on your desk first.

2. STACK & DROP INTO LABELED BAYS:
Follow the Top-Down Box Map on the left. If a tower looks too tall or too short, a block is in the wrong bay—swap it until everything levels!

3. TUCK THIS CARD IN THE LID & SNAP SHUT:
Fold this guide back to [01] and rest it right on top before closing.

12-SOCKET COMPARISON STAGE LID

- Twelve comparison sockets on the Lid Tray let you test fraction towers side-by-side.
- Snap the Stage Lid directly onto the 13U Box Body to keep all 63 blocks secure.
- Easy to clean: wipe with a damp cloth or rinse in lukewarm soapy water.

SHORT A PIECE AT CLEAN-UP TIME?

First check under notebooks and chair legs! If a piece truly vanished, **see PAGE 08 (bottom-right) for your complete Part ID Reference Table.**

✓ FRAC-01 4-POINT CLEAN-UP CHECKLIST

1. 84MM WATERLINE:	All 12 fraction columns (2×1, 2×1/2, 1/3..1/12) level off at 84.0 mm.
2. CLEAR LID TRAY:	Remove any fraction blocks still seated in the 10 Lid Tray sockets.
3. FOLD CARD TO [01]:	Fold this reference booklet to Panel [01] and lay inside the lid.
4. SNAP LID FLUSH:	Snap the Stage Lid flush onto the 13U box rim for silent carry!

MODULAR CLASSROOM & BOOKSHELF NESTING (42.0 MM PITCH)
Every StackCalc box is sized in exact multiples of 42.0 mm (1×1, 3×1, 3×2). All 9 kit boxes interlock side-by-side in a classroom cart or drawer with zero wasted space, and every embossed box icon matches its calculator operation key!

84MM WATERLINE CHECK • EVERY COLUMN LEVELS AT THE TOP

NEXT: CONNECTED LAB ▼ PAGE 07

SIDE B (PAGES 05–08) • READ TOP ROW FIRST

THEN READ BOTTOM ROW (PAGE 07 → PAGE 08)

THE STACKCALC CONNECTED LAB

How Your Kits & RPN Calculator Stack Work Together

ONE CONNECTED SYSTEM FROM KINDERGARTEN TO CALCULUS

Every StackCalc box uses the same modular grid and unit heights—so pieces from different kits actually fit and work together.

WHERE YOUR BOX FITS IN THE 9-KIT LAB

BASE-10 **Ten-Frames Lab** Early Counting & Regrouping	DOMI-01 **Math Domino Studio** Doubles & 12-Frame Part-Whole	FRAC-01 ◀ YOUR BOX **Fraction Studio** Proportional Towers & 4-Way Faces
PRIME-01 **Prime Factor Studio** Multiply by Stacking Prime Towers	ALGB-01 **Algebra Studio** Quadratic Area Arrays & Factoring	TIME-01 **Planetary Clock** 12:1 Geared Hands & Time Wedges
PROB-01 **Probability Dice** Fair Polyhedrals & 10-Sector Spinner	VOLV-01 **Unit Circle Volvelle** Spin-to-Read Exact Trig Radicals	SC-32-HW **SC-32 RPN Calculator** 4-Register Stack Scientific Engine

WHY THIS MATTERS: ZERO "THROW-AWAY" MATH

In most classrooms, every grade uses a totally unrelated plastic toy. In StackCalc, the 7mm unit block you count with in 1st grade is the exact same height as the 1/12th fraction block and the [2] prime block later on!

1. FRAC-01 BOX LID ICON

FDISP
[FDISP] & [g][SIMP]
EMBOSSED ON BOX LID
Build First → Press Key!

2. SC-6 ELEMENTARY RPN

3. SC-32 SCIENTIFIC RPN

▲ **BOX ICON ↔ CALCULATOR KEY CONTINUITY:**
Every physical box lid carries the exact icon of the calculator key that computes that lesson!

HOW BLOCKS CONNECT TO THE CALCULATOR

SHARED MODULAR FOOTPRINT & HEIGHTS

- Shared 7mm unit pitch: the 1/12th block (7.0mm) matches Prime Block [2] and Time Block :05.
- Recessed sockets swallow the bottom peg so only true heights add.
- All storage boxes nest neatly side-by-side on any desk or shelf.

WHY WE CALL IT "STACKCALC" (THE RPN SECRET)

- Face 4 shows the SC-32 RPN reciprocal key: press [4] [1/x] on the calculator to get 0.25.
- Placing a block on the table = pressing [ENTER] to push onto the stack.
- Snapping two towers together = pressing [+] or [×] with zero parentheses!
- Mirrors the 4 stack levels on the SC-32 calculator: X, Y, Z, and T.

FREE ACTIVITY DECKS & INTERACTIVE SIMULATORS

Download extra printable puzzle cards, teacher lesson guides, and interactive browser simulations anytime—no login needed.

Explore All 9 Kits & Free Guides at stackcalc.io

LIVE 4-REGISTER RPN STACK

T: 0 (Top Memory)
Z: 0 (3rd Level)
Y: 0 (1st Block)
X: 3/4 (63.0 mm) (Active Display)

HOW BLOCKS MAP TO STACK KEYS

1. PLACE 1ST PIECE ON STAGE:
Type number + [ENTER] → lifts to Y

2. ADD / STACK 2ND PIECE:
Type 2nd number in X, then [+]/[×]

3. PRESS BOX ICON KEY:
Press [FDISP] & [g][SIMP]
✓ **Zero Parentheses • Always Visible!**

FDISP
BOX ICON = [FDISP] & [g][SIMP] • VERIFIED LESSON SEQUENCES:
- Lesson 1: 4 [ENTER] 6 [÷] 1 [ENTER] 12 [÷] [+] [FDIS → X = 3/4
- Lesson 2: 1 [ENTER] 3 [÷] 1 [ENTER] 4 [÷] [+] [FDISP → X = 7/12

MODULAR 9-KIT CONTINUUM • FROM EARLY COUNTING TO AP CALCULUS

NEXT: PART ID TABLE → PAGE 08

KEEP THIS CARD! PART IDs & VIDEO

Complete 65-Piece Reference & 60-Second 3D Video Guide

PART ID REFERENCE TABLE (65 TOTAL PIECES)

✓ KEEP THIS CARD INSIDE YOUR BOX LID!

Backpacks eat blocks. Dogs chew dice. Vacuum cleaners take no prisoners. Keep this card in your lid so you always have the exact Part IDs below for single-piece refills!

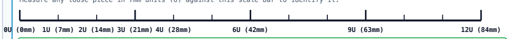
PART NAME	QTY	COLOR / FINISH	PART ID
1.0 Whole Blocks (84mm)	2x	Tan PLA	FRAC-FRAC-01W
1/2 Half Blocks (42mm)	4x	Tan PLA	FRAC-FRAC-02H
1/3 Third Blocks (28mm)	3x	Muted Red	FRAC-FRAC-03T
1/4 Fourth Blocks (21mm)	4x	Matte Blue	FRAC-FRAC-04F
1/5 Fifth Blocks (16.8mm)	5x	White PLA	FRAC-FRAC-05F
1/6 Sixth Blocks (14mm)	6x	Muted Red	FRAC-FRAC-06S
1/8 Eighth Blocks (10.5mm)	8x	Matte Blue	FRAC-FRAC-08E
1/9 Ninth Blocks (9.33mm)	9x	Muted Red	FRAC-FRAC-09N
1/10 Tenth Blocks (8.4mm)	10x	White PLA	FRAC-FRAC-10T
1/12 Twelfth Blocks (7mm)	12x	Matte Blue	FRAC-FRAC-12T

NEED A REPLACEMENT PIECE? NO NEED TO BUY A WHOLE NEW BOX.
Enter any Part ID above at **stackcalc.io/parts** to order individual replacement pieces.

COLOR & PIECE IDENTIFICATION KEY MATCH ANY LOOSE PART

Tan PLA	Muted Red	Matte Blue	White PLA
Base-2: [1] (2x), [1/2] (4	Base-3: [1/3], [1/6], [1/9	Base-4: [1/4], [1/8], [1/1	Base-5: [1/5], [1/10], Lid

UNIVERSAL STACKCALC HEIGHT PITCH (1U = 7.00 MM • 12U = 84.00 MM ONE WHOLE)
Measure any loose piece in 7mm units (U) against this scale bar to identify it:



✓ DISTRICT PO & SAFETY: 10-Pack SKU FRAC-01-CLASS10 • CPSIA / ASTM F963 Non-Toxic Washable PLA/ABS

WATCH THE 60-SECOND 3D VIDEO GUIDE

POINT CAMERA TO WATCH DEMO
stackcalc.io/products/fraction-studio/
See animated 3D examples & pack-up walkthrough

QUESTIONS, CLASSROOM SETS & SUPPORT

- Email us anytime: support@stackcalc.io
- School & District POs: school-orders@stackcalc.io

Designed in Mountain View, CA • Non-Toxic & Washable

COMPANION MANUAL CHAPTERS, STANDARDS & KEYSTROKES

BOOK CHAPTERS: SC-6 Guide Ch. 4 (Fraction Towers) • SC-32 Manual Ch. 3
STANDARDS: CCSS 3.NF.A.3, 4.NF.A.1, 5.NF.A.1 • TEKS 4.3C, 5.3H (Equivalence & LCD)
WATERLINE SPEC: 84.0 mm Waterline (12 Harmonic Fraction Lanes)

ALL 3 VERIFIED CALCULATOR LESSON SEQUENCES (SC6):
Ex 1: 4 [ENTER] 6 [÷] 1 [ENTER] 12 [÷] [+] [FDISP] → X=3/4
Ex 2: 1 [ENTER] 3 [÷] 1 [ENTER] 4 [÷] [+] [FDISP] → X=7/12
Ex 3: 1 [ENTER] 2 [÷] 1 [ENTER] 4 [÷] [+] 1 [ENTER] → X=1

FDISP
BOX ICON KEY: [FDISP] & [g][SIMP]
✓ Cross-Referenced to Companion Book & Calculator Engine
Fold back to [01] & keep this reference card inside your box lid!

LOST A PIECE? GRAB SINGLE REPLACEMENTS ANYTIME AT stackcalc.io/parts

✓ KEEP THIS GUIDE IN BOX LID!

◀ PAGE 07 (ECOSYSTEM) → PAGE 08 (KEEP THIS CARD FOR PART IDs & QR) ▶